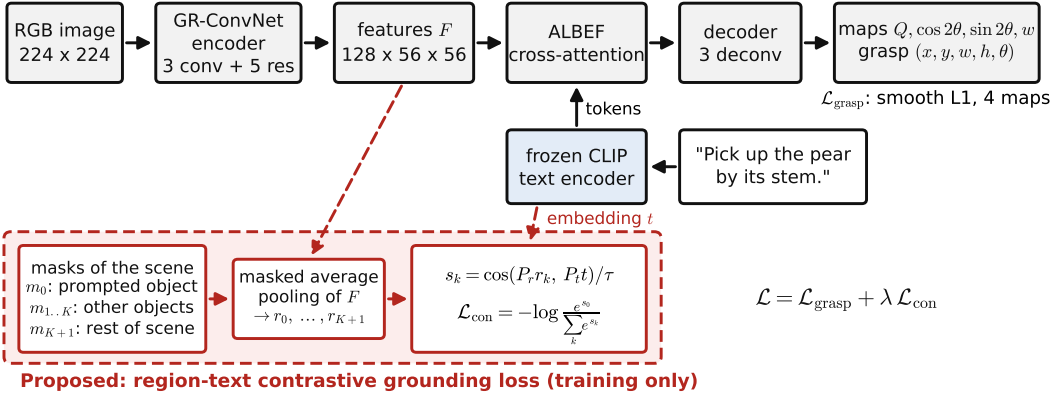
RGB image
224 x 224
GR-ConvNet
encoder
3 conv + 5 res
features F
128 x 56 x 56
ALBEF
cross-attention
decoder
3 deconv
maps $Q, \cos 2\theta, \sin 2\theta, w$
grasp (x, y, w, h, θ)
$\mathcal{L}_{\text{grasp}}$: smooth L1, 4 maps
tokens
frozen CLIP
text encoder
"Pick up the pear
by its stem."
embedding t
masks of the scene
m_0: prompted object
$m_{1..K}$: other objects
m_{K+1}: rest of scene
masked average
pooling of F
$\rightarrow r_0, \ldots, r_{K+1}$
$s_k = \cos(P_r r_k,\ P_t t)/\tau$
$\mathcal{L}_{\text{con}} = -\log \frac{e^{s_0}}{\sum_k e^{s_k}}$
$\mathcal{L} = \mathcal{L}_{\text{grasp}} + \lambda\, \mathcal{L}_{\text{con}}$
Proposed: region-text contrastive grounding loss (training only)